

Decay of the Fourier Transform of Rvachev's Up-Function

Dyadic renormalization, transfer-operator norms, sharp envelopes, and exceptional valleys

Research note for Vladimir Reshetnikov's ProveIt project

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Abstract

Let

$$\Phi(x) = \widehat{\text{up}}(x) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi x}{2^h}\right), \quad \text{sinc } t = \frac{\sin t}{t},$$

be the Fourier transform of Rvachev's compactly supported up-function. The phrase “the decay rate of Φ ” is intrinsically ambiguous: the function vanishes at every nonzero integer, its values are controlled by a lacunary sine product, and even its local extrema have exceptional deep valleys. Consequently there is no global pointwise asymptotic and no single log-normal-power envelope that makes all oscillations approach one nonzero amplitude.

The exact dyadic factorization

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} \prod_{j=1}^n |\sin(\pi 2^j y)|, \quad \frac{1}{2} \leq y \leq 1,$$

reduces every natural averaged notion of decay to a weighted doubling-map problem. We introduce a complete L^q dyadic-shell spectrum. With

$$E_{\kappa}(x) = \exp\left(-\frac{(\log x)^2}{2 \log 2} - \kappa \log x\right),$$

the sharp exponents are

notion of amplitude	κ
geometric mean / almost-everywhere dyadic ray	$\frac{3}{2} + \frac{\log \pi}{\log 2} = 3.151496129 \dots$
arithmetic mean of $ \Phi $	$2.748070014 \dots$
root mean square	$\frac{\log(2\pi)}{\log 2} = 2.651496129 \dots$
uniform upper envelope	$\frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2} = 2.359014879 \dots$

The arithmetic-mean constant is characterized rigorously by the Perron root of an explicit transfer operator and is computed numerically to high precision. The RMS exponent is exact by Parseval, while the uniform exponent follows from an elementary sharp inequality for lacunary sine products; equality is realized by the period-two orbit $1/3 \leftrightarrow 2/3$.

We also prove: (i) every interval between consecutive integer zeros contains exactly one extremum, whose sign is the signed Thue–Morse value; (ii) the normalized arithmetic means have a nonconstant dyadic phase profile, so a plain Cesaro limit does not exist although the weaker two-sided boundedness criterion does hold; and (iii) for each fixed $r \geq 0$, the extremum immediately to the right of $2^{n-1} + r$ satisfies an explicit asymptotic of order $\exp(-(\log x)^2 / \log 2 + \mathcal{O}(\log x))$, twice the generic log-square rate in the leading quadratic term. These results identify precisely what is correct in the existing decay-rate draft: its power $x^{-\log(2\pi)/\log 2}$ is the RMS power, not a pointwise or optimal-envelope power, and no additional universal power of $\log x$ repairs that mismatch.

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1 The question and the shape of the answer

Rvachev's up-function is the even C^∞ probability density supported on $[-1, 1]$ that is closely related to the Fabius distribution function. Under the Fourier convention

$$\widehat{f}(\xi) = \int_{\mathbb{R}} f(t) e^{-2\pi i t \xi} dt,$$

its transform is the entire function

$$\Phi(z) = \widehat{\text{up}}(z) = \prod_{h=0}^{\infty} \text{sinc}\left(\frac{\pi z}{2^h}\right). \quad (1)$$

This formula is classical; see, for example, Arias de Reyna's systematic treatment [1] and the Fabius/Rvachev development in [11].

The motivating questions in [9, 10] ask for the amplitude of the oscillations and, more specifically, for a positive elementary function g such that an average of $|\Phi|/g$ tends to one or at least stays between two positive constants. A working draft in the `ProveIt` repository [12] proposes a log-normal factor together with the power $x^{-\log(2\pi)/\log 2}$ and a possible power of $\log x$. The log-normal factor is indeed fundamental, but the rest of the conclusion depends on which norm or averaging operation is intended.

There are three separate obstructions to a single pointwise answer.

1. The transform has a zero at every nonzero integer, so no asymptotic $|\Phi(x)| \sim g(x) > 0$ can hold on the full real ray.
2. After the deterministic denominator is removed, what remains is a Birkhoff product for the doubling map. Its geometric mean, arithmetic mean, quadratic mean, and supremum have different exponential growth rates.
3. Even the unique maxima between consecutive zeros possess exceptional subsequences. The peaks associated with the binary period-two orbit attain the sharp global upper envelope, whereas the extrema in the first few intervals after a power of two are exponentially smaller on the n^2 scale.

Thus the correct answer is a *spectrum of decay rates*, not one formula. A convenient common scale is

$$E_\kappa(x) := \exp\left(-\frac{(\log x)^2}{2a} - \kappa \log x\right) = x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right), \quad a := \log 2. \quad (2)$$

For fixed dyadic phase, the quadratic term in (2) comes from the product of the n denominators $2^1, \dots, 2^n$. The linear term is where the lacunary numerator enters.

1.1 Main decay table

The following table summarizes the results proved below. In the finite- q rows, “normalized” means the normalized L^q norm on the dyadic shell $[2^{n-1}, 2^n]$.

Amplitude notion	numerator scale Λ	power κ in E_κ	conclusion
Geometric mean ($q = 0$) and typical dyadic ray	$1/2$	$\frac{3}{2} + \frac{\log \pi}{a}$	shell geometric mean is exactly constant after normalization; the same exponent holds for Lebesgue-a.e. fixed phase
Arithmetic mean ($q = 1$)	$\rho_1 = 0.6613226021 \dots$	$\frac{\log \pi + a/2 - \log \rho_1}{a}$	shell means converge to a positive constant; ordinary Cesaro means have a nonconstant dyadic phase profile
Root mean square ($q = 2$)	$2^{-1/2}$	$\frac{\log(2\pi)}{a}$	exact numerator identity $\int B_n^2 = 2^{-n}$; normalized shell RMS converges
Uniform envelope ($q = \infty$)	$\sqrt{3}/2$	$\frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{a}$	global upper bound, sharp on $x = 2^n(2/3)$; not a two-sided envelope for every extremum

The ordering is

$$\begin{aligned} \kappa_\infty = 2.359014879 \dots < \kappa_2 = 2.651496129 \dots \\ < \kappa_1 = 2.748070014 \dots < \kappa_0 = 3.151496129 \dots \end{aligned} \quad (3)$$

Larger q emphasizes rarer and larger values, hence gives a slower decay and a smaller power κ_q .

The coefficient $\log(2\pi)/\log 2$ appearing in the working draft is not an accidental near miss. It is *exactly* the root-mean-square power κ_2 . It is not the arithmetic-mean power, the typical power, or the sharp uniform-envelope power. No factor $(\log x)^p$ can convert one of these powers into another, because their discrepancy is already a nonzero multiple of $\log x$ in the logarithm of the amplitude.

2 Exact products, zeros, and scaling laws

2.1 Entire convergence and three product forms

The compact convergence of (1) follows from

$$\operatorname{sinc} w = 1 - \frac{w^2}{6} + \mathcal{O}(w^4), \quad \sum_{h \geq 0} \frac{|z|^2}{4^h} < \infty$$

uniformly for z in a compact set. Hence Φ is even, real on $\mathbb{R}$, entire, and $\Phi(0) = 1$.

Vieta's product $\operatorname{sinc} t = \prod_{j \geq 1} \cos(t/2^j)$ and Euler's sine product give two other useful forms:

$$\Phi(z) = \prod_{j=1}^{\infty} \cos^j\left(\frac{\pi z}{2^j}\right) = \prod_{m=1}^{\infty} \left(1 - \frac{z^2}{m^2}\right)^{1+\nu_2(m)}. \quad (4)$$

Indeed, in the cosine product the factor with denominator 2^j occurs for exactly j pairs of indices. In the Weierstrass product, the integer m is a zero of the sinc factor at level h precisely when $2^h \mid m$, so its total multiplicity is $1 + \nu_2(m)$.

The elementary one-step scaling law is

$$\Phi(2x) = \text{sinc}(2\pi x)\Phi(x), \quad \Phi(x) = \text{sinc}(\pi x)\Phi(x/2). \quad (5)$$

2.2 Zeros and their multiplicities

Proposition 2.1 (Integer zero divisor). *The nonzero zeros of Φ are exactly the nonzero integers. The zero at $m \in \mathbb{Z} \setminus \{0\}$ has multiplicity*

$$d(m) = 1 + \nu_2(|m|).$$

Moreover, if $b \geq 1$ is an integer, $d = 1 + \nu_2(b)$, and $z \uparrow b$, then

$$|\Phi(z)| = b^{-d} \left| \Phi(b/2^d) \right| (b - z)^d (1 + \mathcal{O}(b - z)). \quad (6)$$

Proof. The divisor statement follows immediately from the last product in (4). For the local coefficient, the vanishing factors are those with $0 \leq h \leq \nu_2(b)$. At $z = b$ each has a simple zero and

$$\left| \frac{d}{dz} \text{sinc}\left(\frac{\pi z}{2^h}\right) \right|_{z=b} = \frac{1}{b}.$$

The product of the remaining factors is

$$\prod_{h=d}^{\infty} \left| \text{sinc}\left(\frac{\pi b}{2^h}\right) \right| = \left| \Phi(b/2^d) \right|.$$

Multiplication gives (6). □

2.3 Exact dyadic renormalization

Let

$$T(x) = 2x \pmod{1}, \quad s(x) = |\sin(\pi x)|, \quad x \in [0, 1],$$

and define the lacunary product

$$B_n(y) := \prod_{j=1}^n s(T^j y) = \prod_{j=1}^n |\sin(\pi 2^j y)|. \quad (7)$$

Theorem 2.2 (Exact dyadic factorization). *For every $n \geq 1$ and $y \in [1/2, 1]$,*

$$|\Phi(2^n y)| = |\Phi(y)| \pi^{-n} y^{-n} 2^{-n(n+1)/2} B_n(y). \quad (8)$$

For any real κ , set

$$W_\kappa(y) := |\Phi(y)| y^\kappa \exp\left(\frac{(\log y)^2}{2a}\right). \quad (9)$$

Then the following normalized identity is exact:

$$\frac{|\Phi(2^n y)|}{E_\kappa(2^n y)} = W_\kappa(y) \exp(n(\kappa a - \log \pi - a/2)) B_n(y). \quad (10)$$

Proof. Iterating (5) gives

$$\Phi(2^n y) = \Phi(y) \prod_{j=1}^n \text{sinc}(\pi 2^j y).$$

The product of the denominators is $\pi^n y^n 2^{1+\dots+n} = \pi^n y^n 2^{n(n+1)/2}$, proving (8). Write $L = \log(2^n y) = na + \log y$ and divide (8) by $\exp(-L^2/(2a) - \kappa L)$. The terms $-n \log y$ cancel, leaving (10). $\square$

Equation (10) is the central identity of this article. The entire deterministic decay has been extracted into E_κ . The bounded phase factor W_κ and the dynamical product B_n contain all remaining information.

2.4 The Thue–Morse polynomial

Let $\tau_k = (-1)^{s_2(k)}$, where $s_2(k)$ is the number of ones in the binary expansion of k . The finite product identity

$$P_n(x) := \prod_{j=0}^{n-1} (1 - e^{2\pi i 2^j x}) = \sum_{k=0}^{2^n-1} \tau_k e^{2\pi i k x} \quad (11)$$

follows by induction. Taking moduli yields

$$|P_n(x)| = 2^n \prod_{j=0}^{n-1} |s(T^j x)|. \quad (12)$$

Thus the same products governing the decay of $\widehat{\text{up}}$ are the normalized moduli of classical Thue–Morse trigonometric polynomials. This is the bridge to Gelfond exponents, Riesz products, and thermodynamic formalism; see [7, 13, 6].

2.5 A second exact scaling identity

The behavior just to the right of a large dyadic integer is exposed by a different exact identity.

Lemma 2.3 (Dyadic-boundary identity). *Let $N = 2^{n-1}$ and $0 < z < N$. Then*

$$|\Phi(N + z)| = \frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \left(\frac{z}{N + z} \right)^n |\Phi(z)|. \quad (13)$$

Proof. For $0 \leq h < n$, the number $N/2^h$ is an integer, so

$$\left| \sin(\pi(N + z)/2^h) \right| = \left| \sin(\pi z/2^h) \right|.$$

The quotient of the first n sinc factors is therefore $(z/(N + z))^n$ in absolute value. Reindexing the remaining tails gives

$$\prod_{h=n}^{\infty} \text{sinc}\left(\frac{\pi(N + z)}{2^h}\right) = \Phi\left(\frac{1}{2} + \frac{z}{2N}\right), \quad \prod_{h=n}^{\infty} \text{sinc}\left(\frac{\pi z}{2^h}\right) = \Phi\left(\frac{z}{2N}\right).$$

Combining the finite and infinite parts proves the identity. $\square$

3 Exactly one extremum per unit interval

The Weierstrass product gives a short complete proof of the observed one-extremum property.

Theorem 3.1 (Strict logarithmic concavity between zeros). *The function Φ is positive and strictly decreasing on $(0, 1)$. For every integer $m \geq 1$, $|\Phi|$ has exactly one critical point $\xi_m \in (m, m+1)$. It is the strict maximum of $|\Phi|$ on that interval. Moreover, for every integer $m \geq 0$,*

$$\operatorname{sgn} \Phi(x) = (-1)^{s_2(m)}, \quad m < x < m+1. \quad (14)$$

For $m \geq 1$ the signed extremum is therefore $\Phi(\xi_m) = (-1)^{s_2(m)} M_m$, where

$$M_m := \max_{m \leq x \leq m+1} |\Phi(x)|.$$

Proof. Away from the integer zeros, termwise logarithmic differentiation of (4) is locally uniform and gives

$$\frac{d}{dx} \log |\Phi(x)| = -2x \sum_{k=1}^{\infty} \frac{1 + \nu_2(k)}{k^2 - x^2}, \quad (15)$$

$$\frac{d^2}{dx^2} \log |\Phi(x)| = -2 \sum_{k=1}^{\infty} (1 + \nu_2(k)) \frac{k^2 + x^2}{(k^2 - x^2)^2} < 0. \quad (16)$$

For $0 < x < 1$, every denominator in (15) is positive, so $(\log \Phi)'(x) < 0$. This proves strict decrease on the first interval.

Now let $m \geq 1$. The logarithmic derivative is strictly decreasing on $(m, m+1)$. The pole contributed by $k = m$ forces it to $+\infty$ at the left endpoint, while the pole contributed by $k = m+1$ forces it to $-\infty$ at the right endpoint. It therefore has exactly one zero, and strict log-concavity makes that point the unique maximum of the modulus.

Starting from $\Phi(0) = 1$, crossing the zero at k changes the sign $1 + \nu_2(k)$ times. Therefore the sign on $(m, m+1)$ is

$$(-1)^{\sum_{k=1}^m (1 + \nu_2(k))}.$$

Legendre's identity $\nu_2(m!) = m - s_2(m)$ gives

$$\sum_{k=1}^m (1 + \nu_2(k)) = m + \nu_2(m!) = 2m - s_2(m),$$

whose parity is $s_2(m)$. This proves (14). $\square$

Remark 3.2. The critical points do *not* in general converge to the midpoints of their unit intervals. In [Section 9](#) we prove that, in every fixed interval offset from a large power of two, the critical point approaches the *right* endpoint at rate $1/n$.

4 Dyadic-shell means and the transfer operator

4.1 Normalized shell means

For $q \in (0, \infty)$ and a measurable H on the shell $D_n = [2^{n-1}, 2^n]$, define

$$\mathcal{M}_{q,n}(H) := \left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} |H(x)|^q dx \right)^{1/q} = \left(2 \int_{1/2}^1 |H(2^n y)|^q dy \right)^{1/q}. \quad (17)$$

The endpoint conventions are

$$\mathcal{M}_{0,n}(H) := \exp\left(\frac{1}{2^{n-1}} \int_{2^{n-1}}^{2^n} \log |H(x)| \, dx\right), \quad (18)$$

$$\mathcal{M}_{\infty,n}(H) := \sup_{2^{n-1} \leq x \leq 2^n} |H(x)|. \quad (19)$$

The logarithm in (18) is locally integrable despite the finitely many zeros.

After $y = (x + 1)/2$, the numerator in (10) becomes

$$\prod_{j=0}^{n-1} s(T^j x), \quad 0 \leq x \leq 1. \quad (20)$$

This product is generated by an explicit Perron–Frobenius operator.

4.2 The weighted doubling operator

For $q > 0$, define

$$(\mathcal{L}_q f)(x) := \frac{1}{2} \left[\sin^q\left(\frac{\pi x}{2}\right) f(x/2) + \cos^q\left(\frac{\pi x}{2}\right) f((x+1)/2) \right]. \quad (21)$$

The factor $1/2$ makes this the Lebesgue Perron operator with weight s^q . For every integrable f ,

$$\int_0^1 f(x) \prod_{j=0}^{n-1} s(T^j x)^q \, dx = \int_0^1 (\mathcal{L}_q^n f)(x) \, dx. \quad (22)$$

We use the following standard Ruelle–Perron–Frobenius fact. The weights in (21) are nonnegative rather than strictly positive at the two endpoints, but the two-branch system is primitive, so the usual cone and Lasota–Yorke proof applies without alteration; see [3, Chs. 2–3] and the singular-potential discussion in [5, 6].

Proposition 4.1 (Perron root and spectral decomposition). *For every $q > 0$, $\mathcal{L}_q$ has a simple leading eigenvalue $\rho_q > 0$, a strictly positive eigenfunction h_q , and a positive eigenmeasure ν_q , which may be normalized by*

$$\mathcal{L}_q h_q = \rho_q h_q, \quad \mathcal{L}_q^* \nu_q = \rho_q \nu_q, \quad \nu_q(h_q) = 1.$$

On a suitable Holder space $C^\alpha[0, 1]$, with $0 < \alpha \leq \min(1, q)$, there is a spectral gap and

$$\rho_q^{-n} \mathcal{L}_q^n f = \nu_q(f) h_q + \mathcal{O}(\vartheta_q^n \|f\|_{C^\alpha}), \quad 0 < \vartheta_q < 1. \quad (23)$$

The same conclusion applies to nonnegative endpoint-power functions encountered below by choosing a compatible Holder exponent.

4.3 The full finite- q decay spectrum

Set

$$\Lambda_q := \rho_q^{1/q}, \quad \kappa_q := \frac{\log \pi + a/2 - \log \Lambda_q}{a}, \quad q > 0. \quad (24)$$

Theorem 4.2 (Sharp finite- q shell asymptotics). *For every $q \in (0, \infty)$ there is a constant $A_q \in (0, \infty)$ such that*

$$\mathcal{M}_{q,n}\left(\frac{\Phi}{E_{\kappa_q}}\right) \longrightarrow A_q \quad (n \rightarrow \infty). \quad (25)$$

More explicitly, with

$$\widetilde{W}_q(x) := W_{\kappa_q} \left(\frac{x+1}{2} \right)^q,$$

and the normalization in [Theorem 4.1](#),

$$A_q^q = \left(\int_0^1 h_q(x) dx \right) \nu_q(\widetilde{W}_q). \quad (26)$$

The number Λ_q is nondecreasing in q , and therefore κ_q is nonincreasing.

Proof. Raise (10) to the q th power, use (17), and substitute $x = 2y - 1$. This gives

$$\begin{aligned} \mathcal{M}_{q,n} \left(\frac{\Phi}{E_\kappa} \right)^q &= \exp(qn(\kappa a - \log \pi - a/2)) \\ &\times \int_0^1 W_\kappa \left(\frac{x+1}{2} \right)^q \prod_{j=0}^{n-1} s(T^j x)^q dx. \end{aligned}$$

By (22) and (23), the integral equals

$$\rho_q^n \left(\int_0^1 h_q \right) \nu_q(W_\kappa((\cdot + 1)/2)^q) (1 + o(1)).$$

The definition (24) is exactly the cancellation condition

$$q(\kappa_q a - \log \pi - a/2) + \log \rho_q = 0.$$

This proves (25) and (26).

For each n , the L^q norm of the product (20) on the probability space $[0, 1]$ is nondecreasing in q . Taking n th roots and then the limit proves that Λ_q is nondecreasing. Formula (24) then gives the opposite monotonicity of κ_q . $\square$

Remark 4.3 (No logarithmic-power correction). For a fixed $q \in (0, \infty)$, replacing E_{κ_q} by $E_{\kappa_q}(\log x)^p$ multiplies the normalized shell mean by $(na)^{-p}(1 + o(1))$. It therefore tends to 0 when $p > 0$ and to $+\infty$ when $p < 0$. At the sharp power, the required logarithmic exponent is $p = 0$.

5 Geometric-mean and typical decay

The $q = 0$ endpoint is especially simple because Lebesgue measure is invariant under the doubling map and

$$\int_0^1 \log \sin(\pi x) dx = -\log 2 = -a. \quad (27)$$

Theorem 5.1 (Exact shell geometric mean). *Define*

$$\kappa_0 := \frac{3}{2} + \frac{\log \pi}{a} = 3.151496129472319 \dots \quad (28)$$

Then

$$\mathcal{M}_{0,n} \left(\frac{\Phi}{E_{\kappa_0}} \right) = \exp \left(\int_0^1 \log W_{\kappa_0} \left(\frac{x+1}{2} \right) dx \right) \quad (29)$$

for every $n \geq 1$. In particular, the normalized shell geometric mean is not merely asymptotically constant; it is exactly independent of the shell.

Proof. Take logarithms in (10), substitute $x = 2y - 1$, and average. The n logarithmic sine terms each have integral $-a$ by invariance and (27). For (28),

$$\kappa_0 a - \log \pi - a/2 = a,$$

so the linear contribution na cancels the integral $-na$ exactly. $\square$

The same exponent governs almost every fixed dyadic phase.

Corollary 5.2 (Almost-everywhere dyadic-ray decay). *For Lebesgue-a.e. $y \in [1/2, 1]$,*

$$\log |\Phi(2^n y)| = -\frac{(\log(2^n y))^2}{2a} - \kappa_0 \log(2^n y) + o(n). \quad (30)$$

Equivalently,

$$|\Phi(2^n y)| = E_{\kappa_0}(2^n y) e^{o(n)}.$$

Proof. The function $\log s$ is integrable and the doubling map is ergodic. Birkhoff's theorem therefore gives

$$\frac{1}{n} \log B_n(y) = \frac{1}{n} \sum_{j=1}^n \log s(T^j y) \longrightarrow -a$$

for almost every y . Substitute this in (10) with $\kappa = \kappa_0$. $\square$

The $o(n)$ in (30) is genuinely dynamical; it need not converge to a constant. Thus even the “typical” statement is a logarithmic asymptotic, not a pointwise equivalent with one universal prefactor.

6 The exact RMS power

For $q = 2$, the transfer operator simplifies dramatically:

$$\mathcal{L}_2 \mathbf{1} = \frac{1}{2} \left(\sin^2 \frac{\pi x}{2} + \cos^2 \frac{\pi x}{2} \right) = \frac{1}{2} \mathbf{1}. \quad (31)$$

Its operator norm on $C[0, 1]$ is at most $1/2$, so

$$\rho_2 = \frac{1}{2}, \quad \Lambda_2 = \frac{1}{\sqrt{2}}. \quad (32)$$

There is also a completely finite Fourier proof. Parseval applied to (11) gives

$$\int_0^1 |P_n(x)|^2 dx = 2^n,$$

while (12) gives

$$\int_0^1 \prod_{j=0}^{n-1} s(T^j x)^2 dx = 2^{-n}. \quad (33)$$

Corollary 6.1 (RMS decay). *The sharp root-mean-square exponent is*

$$\boxed{\kappa_2 = \frac{\log(2\pi)}{\log 2} = 2.651496129472318 \dots} \quad (34)$$

There is a constant $A_2 \in (0, \infty)$ such that

$$\mathcal{M}_{2,n} \left(\frac{\Phi}{E_{\kappa_2}} \right) \longrightarrow A_2. \quad (35)$$

No power of $\log x$ belongs in the leading RMS normalization.

Remark 6.2 (What the draft computed). The power $x^{-\log(2\pi)/\log 2}$ in [12] is precisely $x^{-\kappa_2}$. Thus the draft identified the quadratic-mean scale, albeit without specifying that norm. Its claimed pointwise equivalent and its proposed logarithmic correction do not follow and are incompatible with the exact shell identity (33), the integer zeros, and the exceptional extrema proved later.

7 Arithmetic-mean decay and the Cesaro question

7.1 The Perron root ρ_1

For $q = 1$,

$$(\mathcal{L}_1 f)(x) = \frac{1}{2} \left[\sin \frac{\pi x}{2} f(x/2) + \cos \frac{\pi x}{2} f((x+1)/2) \right]. \quad (36)$$

There appears to be no elementary closed form for its Perron root. It is nevertheless an exact, canonical constant, and it is easy to compute stably:

$$\rho_1 = 0.661322602060565 \dots \quad (37)$$

The corresponding power is

$$\kappa_1 = \frac{\log \pi + a/2 - \log \rho_1}{a} = 2.748070014871335 \dots \quad (38)$$

The decimal in (37) is numerical, but the following elementary Collatz–Wielandt certificate provides a rigorous enclosure.

Proposition 7.1 (A rational enclosure for ρ_1). *One has*

$$0.66126807 < \rho_1 < 0.66134891, \quad (39)$$

and consequently

$$2.74801262 < \kappa_1 < 2.74818899. \quad (40)$$

Proof. For a positive continuous test function f , the Collatz–Wielandt inequalities give

$$\inf_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)} \leq \rho_1 \leq \sup_{x \in [0,1]} \frac{\mathcal{L}_1 f(x)}{f(x)}.$$

Take

$$f(x) = F(\sin \pi x), \quad F(u) = 1 + \frac{376189}{10^6} u - \frac{69093}{10^6} u^2 + \frac{15483}{10^6} u^3.$$

This function is positive on $[0, 1]$. Put $t = \tan(\pi x/4) \in [0, 1]$, $u = 2t/(1+t^2)$, and $v = (1-t^2)/(1+t^2)$. Then

$$\frac{\mathcal{L}_1 f(x)}{f(x)} = \frac{uF(u) + vF(v)}{2F(2uv)},$$

a rational function of t . Exact Sturm-sequence checks show that the numerators of its differences from the two rational endpoints in (39) are positive and have no zeros on $[0, 1]$. A compact exact-arithmetic verification is included in [Section A](#). $\square$

7.2 Sharp dyadic-shell mean

By [Theorem 4.2](#), there is $A_1 > 0$ such that

$$2 \int_{1/2}^1 \frac{|\Phi(2^n y)|}{E_{\kappa_1}(2^n y)} dy \longrightarrow A_1. \quad (41)$$

Therefore the choice

$$g_1(x) := A_1 E_{\kappa_1}(x) \quad (42)$$

makes the mean of $|\Phi|/g_1$ on each dyadic shell tend to one. This gives the precise answer to the natural shell version of the question in [\[10\]](#).

It also answers the weaker full-ray boundedness question.

Corollary 7.2 (Two-sided Cesaro boundedness). *For every fixed $t_0 > 0$, there are constants $0 < c < C < \infty$ such that, for all sufficiently large t ,*

$$c < \frac{1}{t - t_0} \int_{t_0}^t \frac{|\Phi(x)|}{E_{\kappa_1}(x)} dx < C. \quad (43)$$

After multiplying E_{κ_1} by A_1 , the same average tends to one along the dyadic endpoints $t = 2^n$.

Proof. Equation (41) says that the integral over the shell $[2^{n-1}, 2^n]$ is asymptotic to $A_1 2^{n-1}$. Summing the geometric sequence of completed shells proves the dyadic-endpoint limit. If t lies inside the next shell, the completed shells already contribute a positive constant times t , while the entire current shell contributes at most another constant times t . This gives (43). $\square$

7.3 Why the ordinary Cesaro limit has a dyadic phase

The full limit in the stronger condition of [10] does not follow from the shell limit. In fact, the natural normalization (42) has a nonconstant phase profile. This can be stated exactly in terms of the eigenmeasure of $\mathcal{L}_1$.

Normalize h_1, ν_1 as in Theorem 4.1, and put

$$f_1(x) := W_{\kappa_1} \left(\frac{x+1}{2} \right), \quad \mathcal{F}(z) := \frac{\nu_1(f_1 \mathbf{1}_{[0,z]})}{\nu_1(f_1)}, \quad 0 \leq z \leq 1.$$

The measure $f_1 d\nu_1$ is non-atomic, so $\mathcal{F}$ is continuous. Here is a short verification. The endpoint atoms of ν_1 vanish because the corresponding transfer weights vanish. For $z \in (0, 1) \setminus \{1/2\}$, the eigenmeasure equation gives

$$\rho_1 \nu_1(\{z\}) = \frac{1}{2} s(z) \nu_1(\{Tz\}).$$

(The possible atom at $1/2$ is fed only by the already vanishing endpoint atoms.) Hence any remaining atom would generate forward atoms whose masses grow by at least the factor $2\rho_1 > 1$ at every step. A periodic orbit gives an immediate contradiction, and a nonperiodic orbit would produce masses exceeding the finite total mass. Thus ν_1 , and therefore $f_1 d\nu_1$, has no atoms.

Theorem 7.3 (Dyadic phase limit). *For every $y \in [1/2, 1]$ and every fixed $t_0 > 0$,*

$$\lim_{n \rightarrow \infty} \frac{1}{2^n y - t_0} \int_{t_0}^{2^n y} \frac{|\Phi(x)|}{A_1 E_{\kappa_1}(x)} dx = \mathcal{P}(y) := \frac{1 + \mathcal{F}(2y - 1)}{2y}. \quad (44)$$

The continuous function $\mathcal{P}$ satisfies $\mathcal{P}(1/2) = \mathcal{P}(1) = 1$ but is not identically one. Consequently the ordinary Cesaro limit for the natural elementary scale $A_1 E_{\kappa_1}$ does not exist.

Proof. The spectral decomposition gives, for $z \in [0, 1]$,

$$\rho_1^{-n} \int_0^z f_1(x) \prod_{j=0}^{n-1} s(T^j x) dx \longrightarrow \left(\int_0^1 h_1 \right) \nu_1(f_1 \mathbf{1}_{[0,z]}).$$

Approximation from above and below by Holder functions justifies the indicator; the leading conformal measure has no atoms. Returning to the shell variable, the partial integral from 2^{n-1} to $2^n y$ is asymptotic to

$$2^{n-1} A_1 \mathcal{F}(2y - 1).$$

The sum of all preceding complete shells is asymptotic to $2^{n-1} A_1$. Division by $2^n y A_1$ gives (44).

It remains to prove that $\mathcal{F}$ is not the identity distribution. We first note that ν_1 is singular with respect to Lebesgue measure. Otherwise its nonzero absolutely continuous part would yield an L^1 density $r \geq 0$ satisfying

$$\rho_1 r(x) = s(x) r(Tx) \quad \text{for a.e. } x. \quad (45)$$

On the positive invariant support of r , iteration gives

$$r(T^n x) = \frac{\rho_1^n r(x)}{\prod_{j=0}^{n-1} s(T^j x)}.$$

For Lebesgue-a.e. x , Birkhoff's theorem makes the logarithmic growth rate of the right side $\log(2\rho_1) > 0$. On the other hand, for any $r \in L^1$ and any $\epsilon > 0$, Markov's inequality and Borel–Cantelli imply $r(T^n x) \leq e^{\epsilon n}$ eventually for almost every x . Choosing $0 < \epsilon < \log(2\rho_1)$ is a contradiction. Thus ν_1 is singular, and multiplying it by the continuous function f_1 leaves it singular. Hence its normalized distribution $\mathcal{F}$ cannot equal z , the distribution function of Lebesgue measure. But $\mathcal{P} \equiv 1$ would force $1 + \mathcal{F}(2y - 1) = 2y$, or $\mathcal{F}(z) = z$. Therefore $\mathcal{P}$ is nonconstant. $\square$

Remark 7.4. A slowly varying correction, including every power of $\log x$, cannot remove this phase. For a slowly varying L , the ratio $L(2^n y)/L(2^n)$ tends uniformly to one for $y \in [1/2, 1]$, so the profile (44) survives. An exact all-phase normalizer would have to encode the singular conformal distribution, not merely add a smooth logarithmic factor.

8 The sharp uniform envelope

The L^∞ numerator rate can be proved by a sharp elementary inequality. Put

$$\lambda := \frac{\sqrt{3}}{2}. \quad (46)$$

Lemma 8.1 (Sharp two-step logarithmic inequality). *For every $x \in \mathbb{R}$,*

$$\frac{2}{3} \log s(x) + \frac{1}{3} \log s(Tx) \leq \log \lambda, \quad (47)$$

with the usual interpretation at zeros. Equality holds precisely when $x \equiv 1/3$ or $2/3 \pmod{1}$.

Proof. Cubing the desired inequality gives

$$s(x)^2 s(Tx) \leq \lambda^3.$$

With $\theta = \pi x$ and $u = \sin^2 \theta$,

$$s(x)^2 s(Tx) = 2 |\sin^3 \theta \cos \theta|,$$

whose square is $4u^3(1-u)$. The maximum of $u^3(1-u)$ on $[0, 1]$ is $27/256$, attained at $u = 3/4$. Therefore the maximum of the original expression is $3\sqrt{3}/8 = \lambda^3$, with equality at $x \equiv 1/3, 2/3$. $\square$

Proposition 8.2 (Uniform lacunary-product bound). *For every $n \geq 1$,*

$$\lambda^n \leq \sup_{x \in [0,1]} \prod_{j=0}^{n-1} s(T^j x) \leq \lambda^{n-1}. \quad (48)$$

Consequently

$$\lim_{n \rightarrow \infty} \left\| \prod_{j=0}^{n-1} s \circ T^j \right\|_\infty^{1/n} = \lambda = \frac{\sqrt{3}}{2}. \quad (49)$$

Proof. At $x = 1/3$ the orbit alternates between $1/3$ and $2/3$, and every factor equals λ , proving the lower bound. For the upper bound, sum (47) for $x, Tx, \dots, T^{n-2}x$. If $\phi = \log s$, the result is

$$3 \sum_{j=0}^{n-2} \phi(T^j x) - \phi(x) + \phi(T^{n-1}x) \leq 3(n-1) \log \lambda.$$

After adding the final factor,

$$\sum_{j=0}^{n-1} \phi(T^j x) \leq (n-1) \log \lambda + \frac{1}{3} \phi(x) + \frac{2}{3} \phi(T^{n-1} x) \leq (n-1) \log \lambda,$$

because $\phi \leq 0$. Exponentiation gives the upper bound. $\square$

This is the classical Gelfond exponent for the Thue–Morse polynomial: by (12), the unnormalized polynomial grows like $3^{n/2}$ in supremum norm. The ergodic-optimization formulation and its generalizations are developed in [6].

Theorem 8.3 (Sharp global upper envelope). *Define*

$$\kappa_\infty := \frac{\log \pi + a/2 - \log(\sqrt{3}/2)}{a} = \frac{3}{2} + \frac{\log(\pi/\sqrt{3})}{\log 2} = 2.359014879111741 \dots \quad (50)$$

Then there is a constant $C < \infty$ such that

$$|\Phi(x)| \leq C E_{\kappa_\infty}(x), \quad x \geq 1. \quad (51)$$

This exponent is sharp. In fact, for every $n \geq 0$,

$$|\Phi(2^n \cdot 2/3)| = C_* E_{\kappa_\infty}(2^n \cdot 2/3), \quad (52)$$

where

$$C_* = W_{\kappa_\infty}(2/3) = 0.139129774734829 \dots > 0. \quad (53)$$

Proof. In (10), the definition of κ_∞ makes

$$\exp(n(\kappa_\infty a - \log \pi - a/2)) = \lambda^{-n}.$$

After the substitution $x = Ty$, the product is of the form bounded in (48); therefore

$$\frac{|\Phi(2^n y)|}{E_{\kappa_\infty}(2^n y)} \leq \lambda^{-1} \max_{1/2 \leq y \leq 1} W_{\kappa_\infty}(y).$$

This proves the uniform upper bound. At $y = 2/3$, the orbit alternates between $2/3$ and $1/3$, so $B_n(2/3) = \lambda^n$ exactly. Equation (10) then reduces to (52). $\square$

Corollary 8.4 (Sharp smoothness class). *For every $M > 0$, $\Phi(x) = \mathcal{O}(x^{-M})$ as $x \rightarrow \infty$. On the other hand, for every $c > 0$, $e^{cx} |\Phi(x)|$ is unbounded. Thus the transform decays faster than every power but not at any fixed exponential rate.*

Proof. The first statement follows from (51), because the negative quadratic in $\log x$ dominates $-M \log x$. The exact peak ray (52) decays only like $\exp(-\mathcal{O}((\log x)^2))$, which is much larger than e^{-cx} along that subsequence. $\square$

9 Exceptional extrema near dyadic boundaries

The sharp envelope in Theorem 8.3 describes the largest possible values in a shell, but it does not make every local extremum comparable to one constant. The deepest and most explicit obstruction occurs immediately after powers of two.

Theorem 9.1 (Dyadic valleys: exact extremal asymptotic). *Fix an integer $r \geq 0$, and put*

$$b := r + 1, \quad d := 1 + \nu_2(b), \quad N := 2^{n-1}.$$

Let ξ_{N+r} be the unique critical point in $(N + r, N + r + 1)$ and $M_{N+r} = |\Phi(\xi_{N+r})|$. Then, as $n \rightarrow \infty$,

$$\xi_{N+r} = N + b - \frac{db}{n} + o(n^{-1}), \quad (54)$$

$$M_{N+r} \sim \left| \Phi(1/2) \Phi(b/2^d) \right| \left(\frac{b}{N} \right)^n \left(\frac{d}{en} \right)^d. \quad (55)$$

In particular,

$$\log M_{N+r} = -\frac{(\log N)^2}{\log 2} + \mathcal{O}(\log N). \quad (56)$$

The leading quadratic coefficient in (56) is twice the coefficient in E_κ .

Proof. Write $x = N + z$, $r < z < b$, in the exact boundary identity (13). Uniformly for $z \in [r, b]$,

$$\frac{|\Phi(\frac{1}{2} + z/(2N))|}{|\Phi(z/(2N))|} \rightarrow |\Phi(1/2)|, \quad \left(1 + \frac{z}{N}\right)^{-n} \rightarrow 1, \quad (57)$$

because $n/N \rightarrow 0$. Hence

$$M_{N+r} = |\Phi(1/2)| N^{-n} \max_{r \leq z \leq b} (z^n |\Phi(z)|) (1 + o(1)). \quad (58)$$

By (6), as $z \uparrow b$,

$$|\Phi(z)| = D_b (b - z)^d (1 + \mathcal{O}(b - z)), \quad D_b = b^{-d} \left| \Phi(b/2^d) \right|.$$

Any fixed region $z \leq b - \delta$ is exponentially negligible relative to b^n , so the maximum lies in a shrinking neighborhood of b . Set $u = n(b - z)/b$. Uniformly for bounded u ,

$$z^n |\Phi(z)| = D_b b^{n+d} n^{-d} u^d \left(1 - \frac{u}{n}\right)^n (1 + o(1)) \rightarrow D_b b^{n+d} n^{-d} u^d e^{-u}.$$

The function $u^d e^{-u}$ has its unique maximum at $u = d$, with value $d^d e^{-d}$. Standard localization of the maximum therefore gives

$$b - z_{n,r} \sim \frac{db}{n}$$

and

$$\max z^n |\Phi(z)| \sim D_b b^{n+d} n^{-d} d^d e^{-d}.$$

Insert $D_b = b^{-d} \left| \Phi(b/2^d) \right|$ into (58) to obtain (55); the location statement is (54). Finally, $-n \log N = -(\log N)^2/a + \mathcal{O}(\log N)$, proving (56). $\square$

For example, in the first interval after $N = 2^{n-1}$, $r = 0$, $b = d = 1$, and

$$M_N \sim \frac{\Phi(1/2)^2}{en} N^{-n}, \quad \xi_N = N + 1 - \frac{1}{n} + o(n^{-1}). \quad (59)$$

Numerically, $\Phi(1/2) = 0.553771275888810 \dots$

9.1 Consequences for the sequence of extrema

Corollary 9.2 (The upper envelope is sharp but not two-sided). *For $M_m = \max_{[m, m+1]} |\Phi|$,*

$$0 < \limsup_{m \rightarrow \infty} \frac{M_m}{E_{\kappa_\infty}(m)} < \infty, \quad \liminf_{m \rightarrow \infty} \frac{M_m}{E_{\kappa_\infty}(m)} = 0. \quad (60)$$

Therefore no constant multiple of the sharp uniform envelope makes all local extrema approach one nonzero amplitude.

Proof. The upper bound follows from (51). Let $x_n = 2^n(2/3)$ and $m_n = \lfloor x_n \rfloor$. The local maximum in the containing unit interval is at least $|\Phi(x_n)|$, and (52) gives a positive lower bound after division by $E_{\kappa_\infty}(m_n)$; changing the argument by less than one changes E_{κ_∞} by a factor tending to one. This proves the positive limsup.

For the liminf, take $m = N + r$ with fixed r . Comparing (56) with $\log E_{\kappa_\infty}(N) = -(\log N)^2/(2a) + \mathcal{O}(\log N)$ shows that the quotient tends to zero. $\square$

This is a stronger obstruction than the integer zeros alone: even after replacing the function by the absolute values of its unique local extrema, a constant-amplitude normalization does not exist.

10 A continuum of intermediate boundary rates

The boundary identity also explains why many intermediate effective decay rates appear. Suppose $N = 2^{n-1}$, $z = z_n = o(N)$, and $z_n \rightarrow \infty$. From (13), provided the tail ratio remains bounded away from zero and infinity,

$$\log |\Phi(N + z_n)| = \log |\Phi(z_n)| + n \log \frac{z_n}{N} + o(n). \quad (61)$$

If $\log z_n \sim \alpha \log N$ with $0 \leq \alpha < 1$ and z_n itself follows a typical fixed-phase decay law, then substitution of (30) into (61) gives the quadratic coefficient

$$\log |\Phi(N + z_n)| = -\frac{1 - \alpha + \alpha^2/2}{a} (\log N)^2 + \mathcal{O}(\log N) + o((\log N)^2). \quad (62)$$

The coefficient decreases continuously from $1/a$ at $\alpha = 0$ to $1/(2a)$ as $\alpha \uparrow 1$. This calculation is conditional only on the specified decay law for z_n and makes precise the recursive origin of the broad scatter visible in plots of the extrema.

11 Numerical evaluation of the finite- q spectrum

11.1 Chebyshev collocation

For smooth weights (in particular $q \geq 1$), the Perron root of (21) is efficiently obtained by Chebyshev–Lobatto collocation. Let

$$x_j = \frac{1 - \cos(j\pi/(N-1))}{2}, \quad 0 \leq j < N,$$

interpolate f at the points $x_j/2$ and $(x_j + 1)/2$ by the barycentric formula, and form the matrix representation of $\mathcal{L}_q$. Its leading eigenvalue converges rapidly to ρ_q .

For $q = 1$ the convergence is:

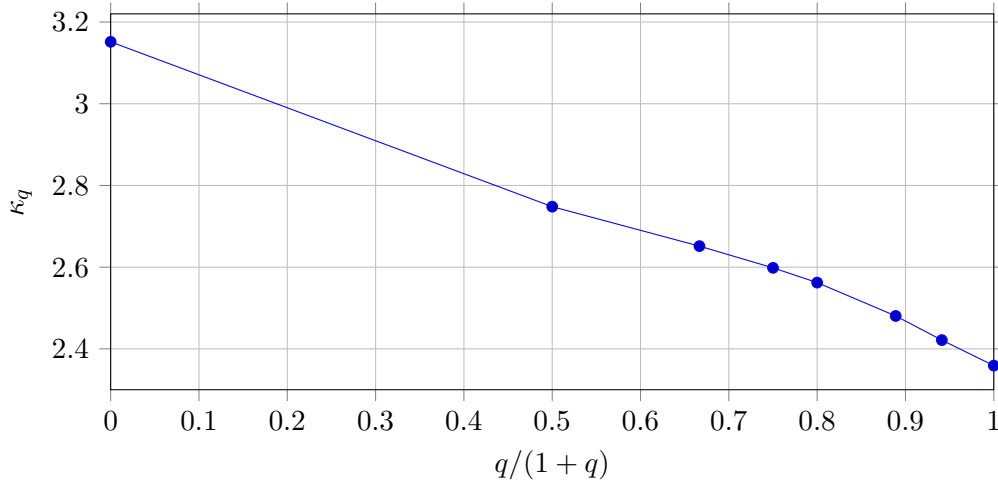
N	leading collocation eigenvalue
8	0.661322639955074
10	0.661322602233874
12	0.661322602061258
16	0.661322602060566
24	0.661322602060565

The rational enclosure in [Theorem 7.1](#) is independent of this numerical convergence and provides a rigorous check on the computed value.

11.2 Sample spectrum

The following values illustrate the interpolation from typical behavior to the sharp uniform envelope. The rows $q = 0, 2, \infty$ are exact; the remaining rows are numerical collocation values.

q	ρ_q	$\Lambda_q = \rho_q^{1/q}$	κ_q
0	–	0.5000000000	3.1514961295
1	0.6613226021	0.6613226021	2.7480700149
2	0.5000000000	0.7071067812	2.6514961295
3	0.3948966324	0.7336593838	2.5983138062
4	0.3201941016	0.7522346457	2.5622414703
8	0.1612444589	0.7960412993	2.4805809434
16	0.0500719357	0.8293247927	2.4214870020
∞	–	0.8660254038	2.3590148791



The monotone curve is not a cosmetic refinement: choosing one row answers one specific notion of “amplitude.” Moving from $q = 0$ to $q = \infty$ changes the power of x by almost 0.8, an effect much larger than any logarithmic correction.

12 Correction of the working draft

We can now state precisely which assertions in [\[12\]](#) survive and which do not.

12.1 Correct common feature

The factor

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

is the common leading shell scale for every fixed finite q and for the sharp shell supremum. It comes from the deterministic denominator $2^{n(n+1)/2}$ in (8). This establishes rigorously the “log-normal-like” character of the largest and averaged values.

12.2 The power $\log_2(2\pi)$ has a precise meaning

The draft's power

$$x^{-\log(2\pi)/\log 2}$$

is exactly the RMS power, by (34). If the intended amplitude was the average of $|\Phi|$, it must be replaced by $x^{-\kappa_1}$ with $\kappa_1 = 2.748070014 \dots$. If the intended object was the sharp global upper envelope, it must be replaced by $x^{-\kappa_\infty}$ with $\kappa_\infty = 2.359014879 \dots$. If the intended object was a typical fixed dyadic phase, the power is $x^{-\kappa_0}$ with $\kappa_0 = 3.151496129 \dots$.

12.3 There is no global equivalent

A statement of the form

$$|\Phi(x)| \sim C(\log x)^{-p} x^{-\kappa} \exp\left(-\frac{(\log x)^2}{2 \log 2}\right)$$

cannot hold globally, for any $C > 0, p, \kappa$, because Φ vanishes at every integer. It also cannot be repaired by replacing $|\Phi(x)|$ with the sequence of local maxima: [Theorem 9.2](#) gives a positive limsup and zero liminf at the sharp upper scale, while [Theorem 9.1](#) exhibits a subsequence with a different leading quadratic coefficient.

12.4 No universal half-logarithm correction

The term $-\frac{1}{2} \log \log x$ asserted in the draft does not arise in any of the exact shell normalizations above. At the RMS scale, Parseval gives a pure exponential-in- n identity. At the arithmetic-mean scale, the transfer operator gives a simple Perron eigenvalue and a positive constant. At the uniform scale, the period-two orbit gives an exact constant ratio. Multiplication by $(\log x)^p$ destroys these normalizations unless $p = 0$.

A power of n does occur in the exceptional valley asymptotic (55), but its exponent is the integer zero multiplicity $d = 1 + \nu_2(r + 1)$ and it accompanies the much faster factor $(b/N)^n$. It is not a universal correction to the generic shell envelope.

13 Conclusions and further directions

The Fourier transform of Rvachev's up-function has an unusually clean deterministic core and an unusually rich oscillatory numerator. The deterministic core is the log-square factor

$$\exp\left(-\frac{(\log x)^2}{2 \log 2}\right),$$

while the numerator is the modulus of a normalized Thue–Morse polynomial. Once the two are separated by (10), the apparently conflicting numerical observations become compatible:

- typical values decay with power $\kappa_0 = 3.151496129 \dots$;

- arithmetic means decay with power $\kappa_1 = 2.748070014\dots$;
- quadratic means decay with the draft's power $\kappa_2 = \log_2(2\pi) = 2.651496129\dots$;
- the largest values decay with the slower sharp power $\kappa_\infty = 2.359014879\dots$;
- extrema near dyadic boundaries can decay with the stronger leading exponent $-(\log x)^2/\log 2$.

Thus the phrase “the exact decay rate” must always be accompanied by a norm, an averaging operation, or a specified subsequence. For the original weak Cesaro criterion, the sharp answer is E_{κ_1} , up to a constant. For the sharp global upper envelope, the exact answer is E_{κ_∞} . For the local-extremum sequence, there is no two-sided envelope on the common log-normal–power scale: binary dynamics creates both period-two peaks and dyadic valleys.

Several natural continuations remain. The pressure function $q \mapsto \log \rho_q$ gives a multifractal large-deviation spectrum for fixed-phase values. A parallel transfer formalism on dyadic cylinders should describe the statistical distribution of the unit-interval maxima M_m , not merely their global limsup and the explicit valley subsequences proved here. Both developments are well suited to eventual formalization because their analytic input is concentrated in one explicit positive operator and a small collection of exact renormalization identities.

A Exact Collatz–Wielandt certificate for ρ_1

The following SymPy script uses only exact rational arithmetic. It constructs the rational function in the proof of [Theorem 7.1](#); Sturm's theorem, invoked by `count_roots`, certifies that the relevant numerators and denominators have no roots on $[0, 1]$. Positivity at $t = 0$ therefore proves positivity throughout the interval.

```
import sympy as sp

t = sp.symbols("t", real=True)
u = 2*t/(1+t**2)
v = (1-t**2)/(1+t**2)

c = sp.Rational(376189, 10**6)
d = sp.Rational(-69093, 10**6)
e = sp.Rational(15483, 10**6)
F = lambda z: 1 + c*z + d*z**2 + e*z**3

R = sp.cancel((u*F(u) + v*F(v)) / (2*F(2*u*v)))
lo = sp.Rational(66126807, 10**8)
hi = sp.Rational(66134891, 10**8)

for expr in (R - lo, hi - R):
    num, den = map(lambda p: sp.Poly(p, t),
                    sp.together(expr).as_numer_denom())
    assert num.eval(0) > 0 and den.eval(0) > 0
    assert num.count_roots(0, 1) == 0
    assert den.count_roots(0, 1) == 0

print("Certified:", lo, "< rho_1 <", hi)
```

For the high-precision decimal, one may replace the exact certificate by a Chebyshev collocation matrix. The rapid stabilization displayed in [Section 11](#) is typical for this analytic weighted-composition operator.

B Notation cross-reference

Symbol	Meaning
$\Phi = \widehat{\text{up}}$	Fourier transform of Rvachev's up-function
a	$\log 2$
T	doubling map $x \mapsto 2x \pmod{1}$
$s(x)$	$ \sin(\pi x) $
$B_n(y)$	$\prod_{j=1}^n s(T^j y)$
$E_\kappa(x)$	$x^{-\kappa} \exp(-(\log x)^2 / (2 \log 2))$
$W_\kappa(y)$	bounded dyadic phase factor in (10)
$\mathcal{L}_q$	weighted Perron operator (21)
ρ_q	Perron root of $\mathcal{L}_q$
Λ_q	$\rho_q^{1/q}$, the L^q numerator growth per dyadic level
κ_q	sharp power in the normalized L^q shell decay
M_m	unique maximum of $ \Phi $ on $[m, m+1]$

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